

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), DECEMBER 2019

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS
(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{bmatrix}$ (3)
- 2 If 2 is an eigen value of $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, without using its characteristic equation, find the other eigen values. (3)
- 3 If $f(x,y) = xe^{-y} + 5y$ find the slope of $f(x,y)$ in the x-direction at (4,0). (3)
- 4 Show that $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$, where $z = e^x \sin y + e^y \cos x$ (3)
- 5 Find the mass of the square lamina with vertices (0,0) (1,0) (1,1) and (0,1) and density function $x^2 y$ (3)
- 6 Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates. (3)
- 7 Test the convergence of the series $\sum_{k=1}^{\infty} \frac{k}{2k+1}$ (3)
- 8 Check the convergence of $\sum_{k=1}^{\infty} \frac{1}{k^{k/2}}$ (3)
- 9 Find the Taylors series for $f(x) = \cos x$ about $x = \frac{\pi}{2}$ up to third degree terms. (3)
- 10 Find the Fourier half range sine series of $f(x) = e^x$ in $0 < x < 1$ (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Solve the system of equations by Gauss elimination method. (7)

$$x + 2y + 3z = 1$$

$$2x + 3y + 2z = 2$$

$$3x + 3y + 4z = 1$$

- b) Find the eigenvalues and eigenvectors of (7)

$$\begin{bmatrix} 4 & 2 & -2 \\ 2 & 5 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

- 12 a) Find the values of λ and μ for which the system of equations (7)

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = \mu$$

has (i) no solution (ii) a unique solution and (iii) infinite solution

- b) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}. \text{ Also write the diagonal matrix.}$$

Module-II

- 13 a) Let f be a differentiable function of three variables and suppose that (7)

$$w = f(x - y, y - z, z - x), \text{ show that } \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$$

- b) Locate all relative extrema of $f(x, y) = 4xy - y^4 - x^4$ (7)

- 14 a) Find the local linear approximation L to the function $f(x, y) = \sqrt{x^2 + y^2}$ (7)

at the point $P(3, 4)$. Compare the error in approximating f by L at the point $Q(3.04, 3.98)$ with the distance PQ .

- b) The radius and height of a right circular cone are measured with errors of at most 1% and 4%, respectively. Use differentials to approximate the maximum percentage error in the calculated volume. (7)

Module-III

- 15 a) Evaluate $\iint_R y \, dx \, dy$ where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$. (7)
- b) Use double integral to find the area of the region enclosed between the parabola $y = \frac{x^2}{2}$ and the line $y = 2x$. (7)
- 16 a) Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$ by reversing the order of integration (7)
- b) Use triple integrals to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. (7)

Module-IV

- 17 a) Find the general term of the series $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{1 \cdot 3 \cdot 5 \cdot 7} + \dots$ and use the ratio test to show that the series converges. (7)
- b) Test whether the following series is absolutely convergent or conditionally convergent $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$ (7)
- 18 a) Test the convergence of $\frac{x}{1 \cdot 2} + \frac{x^2}{2 \cdot 3} + \frac{x^3}{3 \cdot 4} + \dots + \frac{x^k}{k(k+1)} + \dots$ (7)
- b) Test the convergence of the series $\sum_{k=1}^{\infty} \frac{(k+1)!}{4! k! 4^k}$ (7)

Module-V

- 19 a) Find the Fourier series of periodic function with period 2 which is given below $f(x) = \begin{cases} -x & ; -1 \leq x \leq 0 \\ x & ; 0 \leq x \leq 1 \end{cases}$. Hence prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ (7)
- b) Find the half range cosine series for $f(x) = \begin{cases} kx & 0 \leq x \leq L/2 \\ k(L-x) & L/2 \leq x \leq L \end{cases}$ (7)

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- a) Find the Fourier series of $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ x^2 & 0 < x < \pi \end{cases}$ (7)
- b) Obtain the Fourier series expansion for $f(x) = x^2$, $-\pi < x < \pi$. (7)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular and Supplementary Examination December 2020 (2019 Scheme)

Course Code: MAT101**Course Name: LINEAR ALGEBRA AND CALCULUS****(2019 Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 2 & 4 & -3 \end{bmatrix}$ (3)
- 2 Show that the quadratic form $4x^2 + 12xy + 13y^2$ is positive definite. (3)
- 3 If $z = \sin(y^2 - 4x)$ find the rate of change of z with respect to x at the point $(3, 1)$ with y held fixed. (3)
- 4 Find $\frac{dz}{dt}$ by chain rule, where $z = 3x^2 y^2$, $x = t^4$, $y = t^3$ (3)
- 5 Find the mass of the lamina with density function x^2 which is bounded by $y = x$ and $y = x^2$. (3)
- 6 Evaluate $\iint_R y^2 x \, dA$ over the region $R = \{(x, y), -3 \leq x \leq 2, 0 \leq y \leq 1\}$ (3)
- 7 Test the convergence of the series $\sum_{k=1}^{\infty} \left(\frac{k}{100}\right)^k$ (3)
- 8 Does the series $\sum_{k=1}^{\infty} \left(\frac{-3}{4}\right)^k$ converge? If so, find the sum. (3)
- 9 Find the binomial series for $f(x) = (1+x)^{1/3}$ up to third degree term. (3)
- 10 Find the Maclaurin's series of $f(x) = \log(1+x)$ up to third degree term. (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module-I**

- 11 a) Solve the following linear system of equations using Gauss elimination method. $x + y + z = 6$, $x + 2y - 3z = -4$, $-x - 4y + 9z = 18$ (7)

- b) Find eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} \quad (7)$$

- 12 a) Show that the equations:

$$x + y + z = a, \quad 3x + 4y + 5z = b, \quad 2x + 3y + 4z = c$$

(i) have no solution if $a = b = c = 1$ (7)

(ii) have many solutions if $a = \frac{b}{2} = c = 1$

- b) Find the matrix of transformation that diagonalize the matrix (7)

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}. \text{ Also, find the diagonal matrix.}$$

Module-II

- 13 a) Find the local linear approximation of $\frac{4y}{x+z}$ at $(1,1,1)$ (7)

- b) Find the absolute extrema of the function $f(x, y) = x^2 - 3y^2 - 2x + 6y$ over the square region with vertices $(0,0), (0,2), (2,2)$ and $(2,0)$. (7)

- 14 a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$. (7)

- b) Locate all relative extrema of $f(x, y) = 2xy - x^3 - y^2$ (7)

Module-III

- 15 a) Use double integrals to find the area of the region enclosed between the parabola $2y = x^2$ and the line $y = 2x$ (7)

- b) Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 6$. (7)

- 16 a) Change the order of integration and hence evaluate $\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy dx$ (7)

- b) Evaluate $\iiint_G z dV$, where G is the wedge in the first octant cut off from the cylindrical solid $y^2 + z^2 \leq 1$ and the planes $y = x$ and $x = 0$. (7)

Module-IV

- 17 a) Test the convergence of the series

$$1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \dots$$

(7)

- b) Find the sum of the series
- $\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$

(7)

- 18 a) Test the convergence of (i)
- $\sum_{k=1}^{\infty} \frac{k!}{3! (k-1)! 3^k}$
- (ii)
- $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$

(7)

- b) Test the absolute or conditional convergence of
- $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} k^2}{k^3 + 1}$

(7)

Module-V

- 19 a) Expand into a Fourier series,
- $f(x) = e^{-x}$
- ,
- $0 < x < 2\pi$

(7)

- b) Find the half range cosine series for
- $f(x) = (x-1)^2$
- in
- $0 \leq x \leq 1$
- .

(7)

- 20 a) Find the Fourier series of the function
- $f(x) = |x|$
- in
- $-1 \leq x \leq 1$

(7)

- b) Find the Fourier sine series of
- $f(x) = x \cos x$
- in
- $0 < x < \pi$

(7)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
 B.Tech S1 (Special Improvement) Examination January 2021 (2019 scheme)

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS
 (2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

- 1 Determine the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 3 & -1 \\ 5 & 3 & -2 \end{bmatrix}$ (3)
- 2 What kind of conic section is represented by the quadratic form $7x_1^2 + 6x_1x_2 + 7x_2^2 = 200$ Transform it into canonical form. (3)
- 3 Find the derivative of $w = x^2 + y^2$ with respect to t along the path $x = at^2$, $y = 2at$. (3)
- 4 Let $f(x, y) = \sqrt{3x + 2y}$, find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(2, 5)$. (3)
- 5 Evaluate $\int_0^a \int_0^a \int_0^a (yz + xz + xy) dx dy dz$ (3)
- 6 Use polar co-ordinates to evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{\frac{3}{2}} dy dx$ (3)
- 7 Test the convergence of $\sum_{k=1}^{\infty} \frac{1}{(2k+3)^{17}}$ (3)
- 8 Examine whether the series convergence or not $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ (3)
- 9 Find the Maclaurin series of $\frac{1}{x+1}$ up to third degree term. (3)
- 10 Find the Fourier Half Range sine series of $f(x) = x$ in $0 < x < \pi$. (3)

PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Test for consistency and solve the system of equations

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

$$x - y + z = -1$$

(7)

- b) Find the eigenvalues and eigenvectors of $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$

(7)

- 12 a) For what values of a and b do the system of equations

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + az = b$$

have i) no solution ii) unique solution iii) more than one solution.

(7)

- b) Find the matrix of transformation that diagonalize the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \text{ Also write the diagonal matrix.}$$

Module-II

- 13 a) If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$.

(7)

- b) If the local linear approximation of a function $f(x, y, z) = xy + z^2$ at a point P is $L(x, y, z) = y + 2z - x$, find the point P.

(7)

- 14 a) If $z = e^{xy}$, $x = 2u + v$, $y = \frac{u}{v}$ find $\frac{\partial z}{\partial u}$.

(7)

- b) Locate all relative extrema of $f(x, y) = 3x^2 - 2xy + y^2 - 8y$.

(7)

Module-III

- 15 a) Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by reversing the order of integration.

(7)

- b) Using triple integral find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + 2y + z = 6$.

(7)

- 16 a) Find the mass and center of gravity of the triangular lamina with vertices $(0,0)$, $(0,1)$ and $(1,0)$ and density function $\delta(x,y) = xy$ (7)

- b) Evaluate $\iint_R x^2 dy dx$, where R is the region between $y = x$ and $y = x^2$ (7)

Module-IV

- 17 a) Discuss the convergence of the series (7)

$$(i) \sum_{k=1}^{\infty} \frac{k!}{k^k} \quad (ii) \sum_{k=1}^{\infty} \left(\frac{k}{k+1} \right)^{k^2}$$

- b) Examine the convergence and divergence of the series (7)

$$\frac{x}{1.3} + \frac{x^2}{3.5} + \frac{x^3}{5.7} + \dots$$

- 18 a) Test the convergence of $1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \dots$ (7)

- b) Prove that the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{(k-2)}{k(k+1)}$ is conditionally convergent. (7)

Module-V

- 19 a) Obtain Fourier series for the function $f(x) = |\sin x|$ $-\pi < x < \pi$ (7)

- b) If $f(x) = \begin{cases} kx; & 0 < x < \frac{\pi}{2} \\ k(\pi - x); & \frac{\pi}{2} < x < \pi \end{cases}$ then show that (7)

$$f(x) = \frac{4k}{\pi} \left(\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \dots \right).$$

- 20 a) Find the Fourier cosine series of $f(x) = x^2$ in $(0, \pi)$. Hence show that (7)

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

- b) Find the Fourier series for the function (7)

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 1 - x & 1 < x < 2 \end{cases}$$

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS

(2019 -Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks*

Marks

- 1 Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$ (3)
- 2 Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. What are the Eigen values of A^2 , A^{-1} without using its characteristic equation. (3)
- 3 If $z = \frac{xy}{x^2+y^2}$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$. (3)
- 4 Show that the equation $u(x, t) = \sin(x - ct)$, satisfies wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$. (3)
- 5 Evaluate $\int_0^3 \int_0^2 \int_0^1 xyz \, dx \, dy \, dz$. (3)
- 6 Find the mass of the lamina with density $\delta(x, y) = x + 2y$ is bounded by the x -axis, the line $x = 1$ and the curve $y^2 = x$. (3)
- 7 Find the rational number represented by the repeating decimal 5.373737 ... (3)
- 8 Examine the convergence of $\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3}$ (3)
- 9 Find the Taylor series expansion of $f(x) = \sin \pi x$ about $x = \frac{1}{2}$ (3)
- 10 If $f(x)$ is a periodic function with period 2π defined in $[-\pi, \pi]$. Write the Euler's formulas a_0, a_n, b_n for $f(x)$. (3)

PART B

Answer one full question from each module, each question carries 14 marks.

MODULE 1

- 11 a Solve the following linear system of equations using Gauss elimination method (7)

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

$$2x - 2y + 3z = 2$$

- b Find the eigenvalues and eigenvectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (7)

- 12 a Solve the following linear system of equations using Gauss elimination method. (7)

$$2x - 2y + 4z = 0$$

$$-3x + 3y - 6z + 5w = 15$$

$$x - y + 2z = 0$$

- b Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$. (7)

Also write the diagonal matrix.

MODULE 2

- 13 a The length and width of a rectangle are measured with errors of at most 3% and 4% respectively. Use differentials to approximate the maximum percentage error in the calculated area. (7)

- b Find the local linear approximation L of $f(x, y, z) = xyz$ at the point $P(1, 2, 3)$. Compute the error in approximation f by L at the point $Q(1.001, 2.002, 3.003)$. (7)

- 14 a If $w = f(P, Q, R)$ where $P = x - y$, $Q = y - z$, $R = z - x$ prove that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$. (7)

- b Locate all relative extrema and saddle points of $f(x, y) = 4xy - x^4 - y^4$ (7)

MODULE 3

- 15 a Find the area bounded by the parabolas $y^2 = 4x$ and $x^2 = \frac{y}{2}$. (7)

- b Evaluate $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} dx dy$ using polar coordinates. (7)

- 16 a Evaluate $\int_0^1 \int_y^1 \frac{x}{x^2+y^2} dx dy$ by reversing the order of integration. (7)

- b Use triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. (7)

MODULE 4

- 17 a Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$ (ii) $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ (7)
- b Test whether the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{\sqrt{k+1}}$ is absolutely convergent or conditionally convergent (7)
- 18 a Test the convergence of the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \dots$ (7)
- b Test the convergence of (i) $\sum_{k=1}^{\infty} \left(\frac{k}{k+1}\right)^{k^2}$ (ii) $\sum_{k=1}^{\infty} \frac{7^k}{k!}$ (7)

MODULE 5

- 19 a Find the Fourier series expansion of $f(x) = x - x^2$ in the range $(-1, 1)$. (7)
- b Obtain the half range Fourier cosine series of $f(x) = e^{-x}$ in $0 < x < 2$ (7)
- 20 a Find the Fourier series expansion of $f(x) = x^2$ in the interval $-\pi < x < \pi$. (7)
- Hence show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$
- b Obtain the half range Fourier sine series of $f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases}$ (7)
